TravelTide

For Elena Tarrant, Head of Marketing, and the leadership team

Keeping our customers coming back

Goal: Launch a personalized rewards program by matching every customer with the perk that fits their travel habits.

Question 1

Does the data back your five perks?



Question 2

Which perk for each customer?



One likely favorite perk per customer, so the invitation email can lead with it

Jacklynn Tong · Data Analytics · TravelTide Mastery Project · 6 Nov 2026

What we did

From 1 million users to one perk each

1,020,926
all TravelTide users

979,801
active since Jan 4, 2023

5,998 customers
with more than 7 sessions

83 %
in the USA

41
median age

49,211
sessions

15,489
kept trips

1. **Profile customers**
   Cleaned the data and built a profile per customer from their trips: party size, nights, distance, discounts, repeat cities

2. **Find groups**
   8 groups built around clear habits, such as hotel-only stays. Each one tested: real habit or just luck?

3. **Cross-check**
   Machine learning (K-Means) that never saw our rules found 4 of the same groups on its own

4. **Match perks**
   Each group gets the perk that fits its habit (76 %). The 24 % without a clear habit get their best-scoring perk

Cohort as you suggested: sessions from Jan 4, 2023, users with more than 7 sessions. SQL and Python notebooks on GitHub.

The short answer

Your perks partly hold up, and every customer has one

Question 1: does the data back your five perks?

Partly yes

Backed by a habit
Free hotel meal, no cancellation fees, 1 night free hotel with flight

No clear habit, so test first
Exclusive discounts, free checked bag

New idea from the data
Extra-legroom seat for long-haul travelers

Question 2: which perk for each customer?

Yes, all 5,998



4,561 by a segment rule

1,437 by perk score

9 customer groups share 6 perks
One perk per customer, no overlaps

Saved for every customer
user_perks.csv, ready for the invitation emails

Who gets which perk

Nine groups: who they are and what they get

Dreamers 556
Browse, but no kept trip yet
No cancellation fees
Why: a risk-free first booking

Family Crew 417
Travel with their children
Free hotel meal
Why: real savings on family stays

Couples & Friends 829
Travel together, no children
No cancellation fees
Why: easier to commit as a group

Hotel Stayers 616
>= 50% Hotel-only trips, about 6 nights
Free hotel meal
Why: hotel-only habit, +26 %

Favorite-City Returners 604
Return to the same city
1-night free hotel with flight
Why: rewards the next visit

Deal Hunters 517
Discount on half of their trips
Exclusive discounts (test first)
Why: discount use looks like luck

Long-haul Travelers 377
Fly 3,000 km+, coast to coast
Extra-legroom seat (new)
Why: long flights are a habit, +9 %

Package Travelers 645
Always flight + hotel together
1 night free hotel with flight
Why: package habit, +17 %

All-rounders 1,437
No single habit stands out
Their best-scoring perk
Why: perk score across all five

Groups in the order the rules are checked; the first match decides. 4,561 customers by rule (76 %), 1,437 by perk score (24 %).

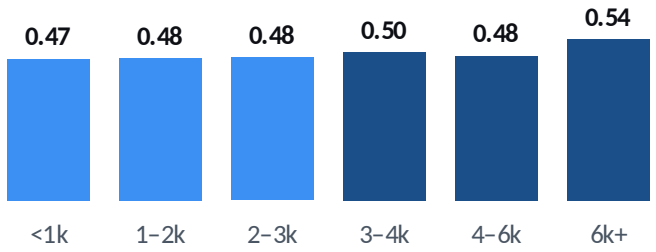
What the data showed early

Three of many findings that shaped our decisions

Finding 1 · flights → rule + perk

Long flights, same luggage

Checked bags per passenger by flight distance (km) · 13,157 kept flights



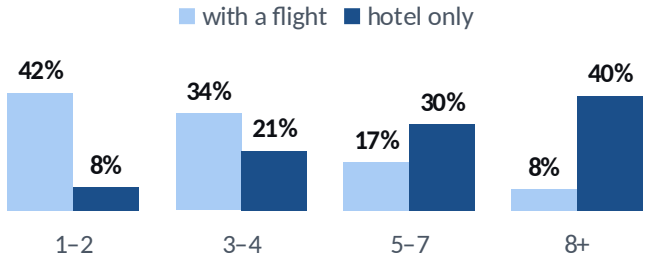
navy = long-haul, 3,000 km+

→ Long-haul Travelers get an extra-legroom seat, not a free bag (the bag becomes a pilot)

Finding 2 · hotels → rule + perk

Hotel-only stays last twice as long

Share of stays by nights · median 6 nights hotel-only vs. 3 with a flight



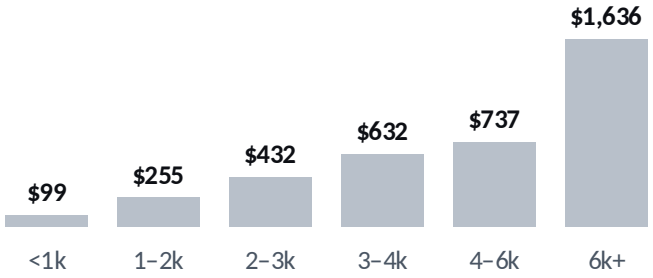
nights per stay

→ Hotel Stayers (half their trips or more hotel-only) get the free hotel meal: more nights, more meals

Finding 3 · fares → feature left out

Fares are just distance

Median fare per passenger by flight distance (km) · r = 0.98 · 13,157 kept flights



about $0.18 per km and passenger

→ Price adds nothing new, so no price-based group: distance alone defines the Long-haul Travelers

Live demo in Tableau

Summary

Key takeaways and next steps

Key insights

- **3 of your 5 perks** match a real habit: hotel stays, packages and first-time bookers
- **Discounts and checked bags** look like luck
- **Long flights** are a habit worth a new perk: the extra-legroom seat

Conclusion

Perks work when they follow how customers already travel.

Every one of the 5,998 customers now has one perk, 76 % by a clear rule, ready in user_perks.csv for the invitations.

Recommendation

- **Launch now:** 4 perks, 4,769 customers (80 %)
- **Test first:** discounts and the free bag, 3 months vs. a control group
- **Pilot:** 48-hour fare lock for the Dreamers
- **Track:** sign-ups, booking rate, trips, cancellations

Decision needed: a go for the four-perk launch and the 3-month test, so the first invitations can go out.

TravelTide

Thank you for your trust

Two things to take away for our customer retention plan:

1

Lead with the right perk

Every customer has one perk. Four are backed by real habits and ready to launch: 4,769 customers, 80 %.

2

Test before you scale

A/B-test exclusive discounts and the free checked bag against a control group: 1,229 customers, 20 %.

Questions?

Jacklynn Tong · Data Analytics · notebooks, SQL and user_perks.csv on GitHub: github.com/jtong28/TravelTide

Backup 1 of 4: data preparation

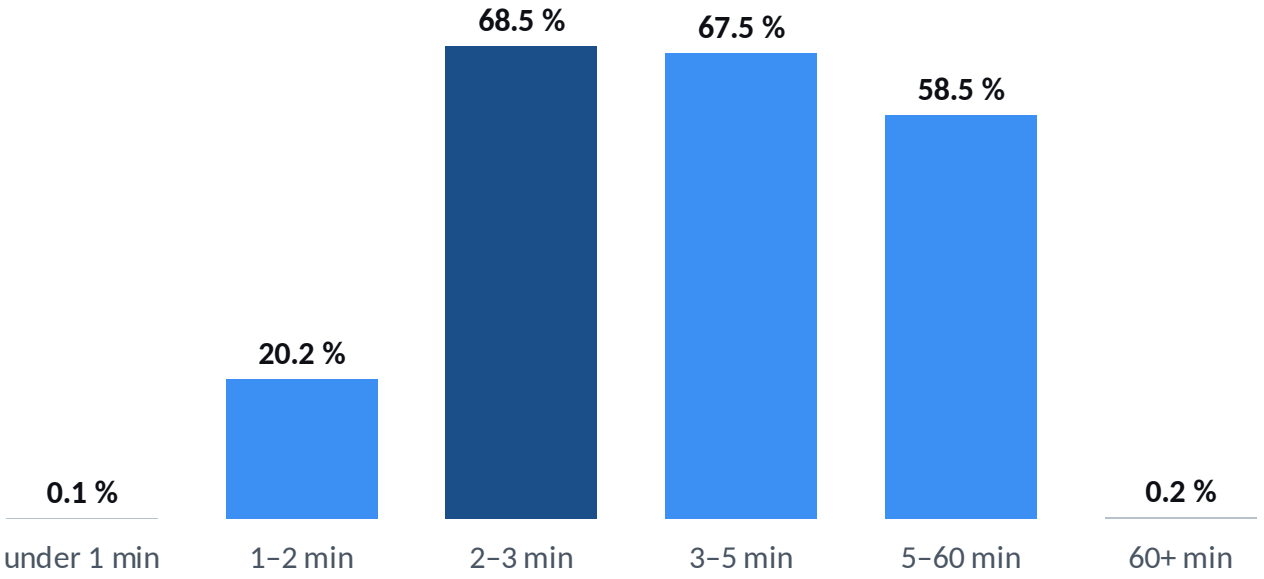
What we fixed before counting anything

Issue	What we did	Effect
Cancellations	A canceled trip is not travel, so its booking doesn't count	610 trips out, 15,489 kept
Hotel nights	Raw nights counted 24-hour periods and had negative values, so we recalculated them from check-in and check-out dates	421 bookings with invalid dates: spend left as unknown
Discount flags	Flags show what was offered, so a discount only counts on the part that was booked	549 trips corrected: 26.3 % → 22.7 % discounted
Birth year 2006	Placeholder birthdates: 233 users, almost all of the under-18s	Flagged; age plays no part in the perks
Time window	7 canceled trips were booked before Jan 4, 2023	Kept, no effect on the kept trips

Details: notebook 01 (EDA and cleaning) and sql/session_and_user_base.sql on GitHub.

Session length: a 1-minute floor and a 120-minute cap

Share of sessions that end in a booking, by session length



49,211 sessions · 60+ min: 464 sessions, 99.8 % of them cancellations

0.1 % — Floor

of sessions under 1 minute book. The shortest booking: 54 s, 7 clicks.

69 % — Sweet spot

book at 2–3 minutes. Longer sessions don't book more.

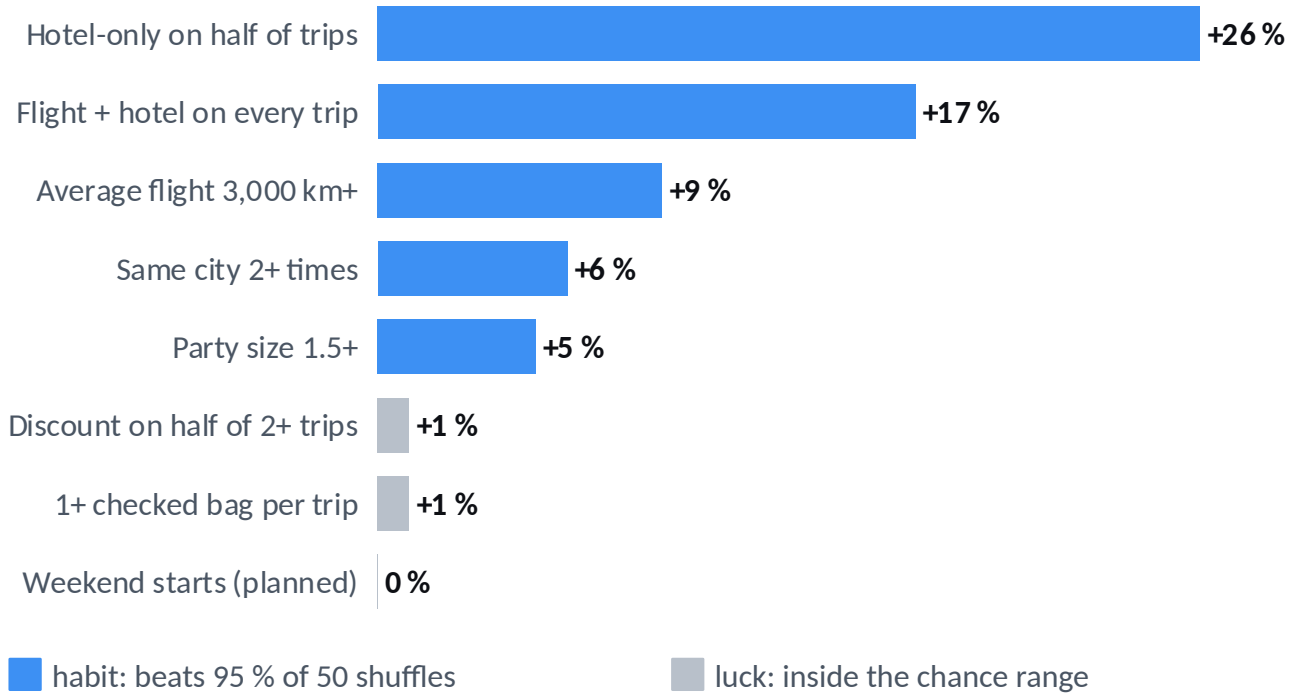
120 min — Cap

and 200 clicks: the hard limit of every cancellation session.

A logging limit, not real browsing, so cancellation sessions stay out of the behavior features. Source: notebook 01, section 4.

Backup 3 of 4: habit or luck

Three checks behind every perk



50 × — Shuffle test

Trips dealt out at random 50 times. 5 rules beat 95 % of the shuffles.

458 vs 236 — Chance model

Customers who never book vs. chance: the Dreamers are a real group.

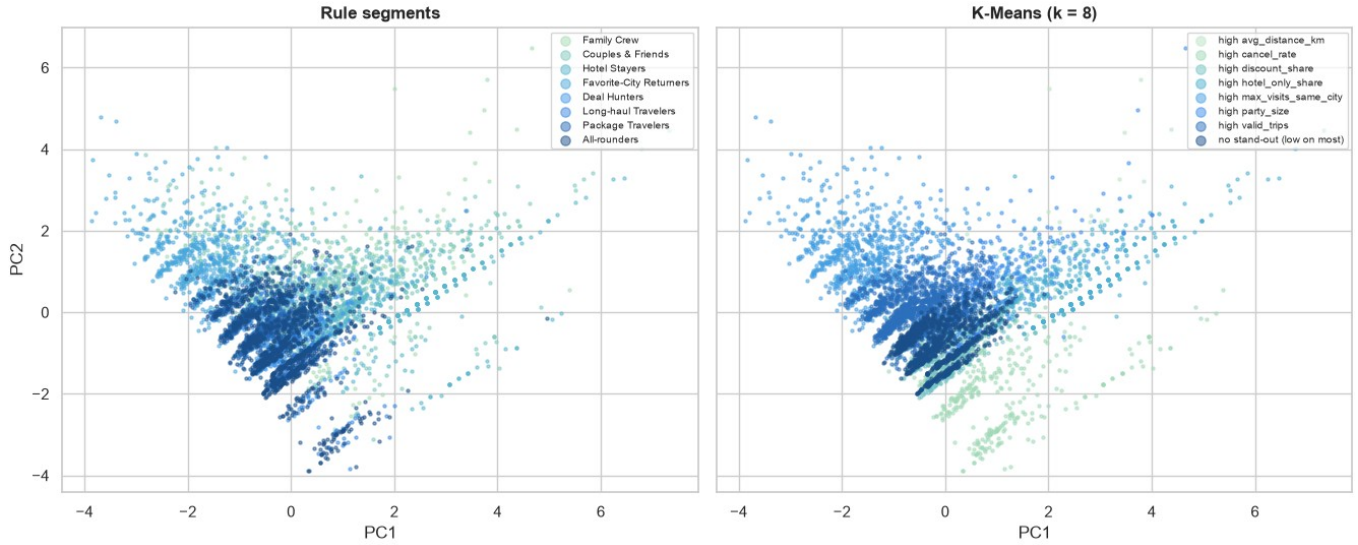
$|r| \leq 0.12$ — Redundancy

Perk scores barely overlap. The hotel perks reach opposite customers.

% = customers above what chance alone gives. Details: notebook 03, sections 3 and 4.1.

Machine learning as a cross-check

Same customers, two views: our rule segments (left) and K-Means clusters (right)



All 5,442 bookers on the first two PCA components. Behavior is one continuum, with a few groups that stand apart at the edges.

4 of 7 — K-Means (k = 8)

booker groups found on its own, most clearly Favorite-City and Hotel Stayers

97 % — DBSCAN

of the 115 most extreme customers flagged as outliers, like our own check

7 of 9 — PCA

components needed for 80 % of the variance, so there's no shortcut

Clustering shows who is similar; the shuffle test shows whether it's a habit. Details: notebook 03, sections 5–7.